PROJECT PROPOSAL

GROUP	:	1	
COURSE	:	CSE299	
SECTION	:	3	
FACULTY	:	MSRB(Dr.M.Shifat-E-Rabbi)	
DEPARTMENT	:	ECE	
GROUP MEMBERS	:	ID	NAME
		2111540642	Nuhash Bin Hassan
		2232755642	Abu Md.Ashraful Rahat
		2311086642	Md.Shakib Rafid Shakib
		2321534042	Muhammad Farhan Rafid

PROJECT PROPOSAL: AI-POWERED E-COMMERCE PLATFORM WITH INTELLIGENT BUSINESS INSIGHTS

1. INTRODUCTORY OVERVIEW

In the rapidly evolving digital marketplace of Bangladesh, platforms like Ghorer Bazar and Alik Food have demonstrated the immense potential of localized e-commerce. However, small to medium enterprises often struggle with the dual burden of managing customer relations and analyzing complex operational data. This project introduces a streamlined, production-ready AI-Powered E-Commerce Platform engineered to bridge this gap without introducing excessive structural bloat.

The system functions as a robust online storefront with fully integrated customer profiles, product catalogs, automated cart dynamics, secure checkout pipelines, and a detailed administration dashboard. To maximize developer efficiency and project feasibility within a single semester, the system isolates its intelligent capabilities into two distinct, open-source AI chatbot interfaces powered by local models via Ollama.

The first chatbot serves as an interactive client assistant that manages consumer queries, guides product discovery, and routes complex issues directly to a human-takeover chat stream. The second chatbot acts as an automated Admin Manager, enabling business owners to query real-time database state, inventory updates, and basic sales performance using natural language.

By utilizing Docker containers for development environment isolation and a structured Git CI/CD workflow, the development team ensures reliable system deployment while tracking code iterations directly through the project's public repository. This project establishes an accessible, intelligent retail architecture that empowers local merchants with automated operational insights and optimized consumer support pathways.

2. FORMAL PROBLEM STATEMENT

Modern retail operations require continuous data analysis and constant customer engagement, creating major technical bottlenecks for emerging digital brands in Bangladesh:

- **Resource Constraints in Customer Relations:** Small e-commerce teams struggle to manage high-volume customer message streams manually, causing delayed responses and cart abandonment.
- **Complex Operational Data Retrieval:** Business managers frequently struggle to parse SQL relational database records to locate low-stock metrics or find top-selling monthly items.
- **Development Environment Drift:** Inconsistent local system dependencies across team members often cause integration errors during code assembly and weekly technical reviews.
- **Disjointed Support Channels:** Standard automated chat frameworks isolate user interactions, preventing human operators from intervening dynamically when consumer negotiations get complicated.

This project directly resolves these issues by delivering a self-contained, containerized e-commerce engine that combines core transactional workflows with accessible local AI support tools.

3. CORE TECHNICAL ARCHITECTURE

The platform is organized into three primary modules designed to manage standard store operations and execute intelligent data analysis safely.

3.1 Customer & Admin Dashboard

- **Customer Interface:** Includes secure registration/login, user profile management, smooth search functionality, and product browsing. It also features standard cart operations, order history logs, and a structured checkout system.
- **Administrative Interface:** Provides management dashboards to configure product items, catalog categories, inventory levels, and transaction states. It also includes an automated utility to generate printable PDF invoices and official sales receipts for completed orders.

3.2 Dual AI Chatbot Infrastructure (using Ollama Framework)

- **AI Customer Support Chatbot:** Uses a locally deployed open-source language model to address consumer product questions and process account messages. It includes a human-takeover layer that smoothly transitions the active session to a manual operator dashboard if the AI model reaches its functional limit.
- **AI Admin Manager Chatbot:** Connects directly to backend storage logs, allowing administrators to review operational status through natural language prompts. It supports direct analytic queries such as "How many orders today?", "Which products are low in stock?", and "Top selling products this month?".

3.3 Relational Database Schema Model

The system relies on a consistent database architecture that isolates transactional fields across seven fundamental logical entities:

I. **Users:** Stores customer profiles, administrative credentials, and operational security tokens.
II. **Products & Categories:** Manages specific item descriptions, pricing data, and organizational catalogs.
III. **Inventory:** Maintains real-time warehouse stock counts, supply limits, and fulfillment indicators.
IV. **Orders & Order Items:** Tracks customer invoices, payment methods, transaction states, and single item metrics.
V. **Chat Messages:** Chronologically logs conversational data exchanged between users, the AI chatbot, and manual support administrators.